BIO504P – BIOCHEMISTRY I

Practical Exam Notes

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SHORT QUESTIONS

1. Differentiate between unsaturated, saturated and super saturated solution.

Unsaturated Solution: An unsaturated solution is a solution that contains less solute than the maximum amount that can dissolve at a given temperature. In this type of solution, additional solute can still dissolve easily without settling at the bottom. For example, if a small amount of salt is dissolved in water and more salt can still be added, the solution is unsaturated.

Saturated Solution: A saturated solution contains the maximum amount of solute that can dissolve in a solvent at a specific temperature. At this stage, the solution cannot dissolve any more solute, and any extra solute added remains undissolved. For example, when sugar is continuously added to water until no more dissolves, the solution becomes saturated.

Supersaturated Solution: A supersaturated solution contains more dissolved solute than a saturated solution at the same temperature. It is usually prepared by heating the solution to dissolve extra solute and then cooling it slowly. This type of solution is unstable, and even a slight disturbance can cause the excess solute to crystallize out of the solution.

2. Differentiate between optimum pH and optimum temperature.

Optimum pH: Optimum pH is the specific pH value at which an enzyme or organism shows maximum activity and functions most efficiently. Each enzyme has its own optimum pH depending on its environment. If the pH becomes too acidic or too basic, the enzyme activity decreases because the structure of the enzyme may change. For example, pepsin works best at pH 2 in the stomach.

Optimum Temperature: Optimum temperature is the specific temperature at which an enzyme or organism performs its activity at the highest rate. At this temperature, molecular movements and enzyme reactions occur efficiently. If the temperature rises too high, the enzyme may denature, while very low temperatures slow down its activity. For example, most human enzymes work best at 37°C.

3. Define saponification.

Saponification is the chemical process in which fats or oils react with a strong alkali, such as sodium hydroxide (NaOH) or potassium hydroxide (KOH), to produce soap and glycerol. It is commonly used in soap manufacturing.

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4. What is use of Benedict's test and write its principle.

Use of Benedict's test:

Benedict's test is used to detect the presence of reducing sugars such as glucose, lactose, and maltose in a solution.

Principle:

The test is based on the reduction of copper (II) sulfate present in Benedict's reagent into copper oxide by reducing sugars in an alkaline medium upon heating. The color changes from blue to green, yellow, orange, or brick-red depending on the amount of reducing sugar present.

5. What do you know about amino acids?

Amino acids are the basic building blocks of proteins. They are organic compounds made up of an amino group ($–NH_2$), a carboxyl group (–COOH), hydrogen, and a variable side chain attached to a central carbon atom. Amino acids combine through peptide bonds to form proteins. They are important for growth, tissue repair, enzyme formation, and many metabolic activities in living organisms. There are about 20 common amino acids found in proteins.

6. Write procedure of Biuret test.

Procedure of Biuret Test:

- ✓ Take about 2 mL of the test solution in a test tube.
- ✓ Add 2 mL of sodium hydroxide (NaOH) solution.
- ✓ Then add a few drops of copper sulfate ($CuSO_4$) solution.
- ✓ Shake the test tube gently and observe the color change.
- ✓ Appearance of a violet or purple color indicates the presence of proteins.

7. How organic solvent and biomolecules move in chromatography?

In chromatography, the organic solvent acts as the mobile phase and moves through the stationary phase, such as paper or silica gel, by capillary action. As the solvent travels upward or across the stationary phase, it carries the dissolved biomolecules with it.

Different biomolecules move at different speeds because of differences in their solubility in the solvent and their attraction toward the stationary phase. Biomolecules that are more soluble in the organic solvent and less strongly attached to the stationary phase move farther and faster. In contrast, molecules that have stronger attraction to the stationary phase move slowly and travel a shorter distance. This difference in movement helps in the separation and identification of biomolecules.

8. Define solute and solvent.

Solute: A solute is the substance that gets dissolved in a solution. It is usually present in a smaller amount. For example, salt in salt water is the solute.

Solvent: A solvent is the substance that dissolves the solute to form a solution. It is usually present in a larger amount. For example, water in salt water is the solvent.

9. What is retardation factor and how it is calculated?

The retardation factor (Rf value) is the ratio that shows how far a substance travels in chromatography compared to the distance traveled by the solvent. It is used to identify and compare different compounds. It is calculated by using the formula:

$$Rf = \frac{Distance\ traveled\ by\ solute}{Distance\ traveled\ by\ solvent}$$

For example, if a pigment moves 4 cm and the solvent front moves 8 cm, then:

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$$Rf = \frac{4}{5}$$

The Rf value is always less than 1.

10. Name three reagents used in Xanthoproteic test.

The three reagents commonly used in the Xanthoproteic test are:

1. Concentrated nitric acid (HNO_3)
2. Sodium hydroxide (NaOH) or ammonia solution
3. Protein solution (sample)

This test is used to detect aromatic amino acids in proteins.

11. Write procedure of Ninhydrin test.

Procedure of Ninhydrin Test:

- ✓ Take 2 mL of the sample solution in a test tube.
- ✓ Add a few drops of ninhydrin reagent to the solution.
- ✓ Heat the test tube gently in a water bath for 1–2 minutes.
- ✓ Allow the solution to cool.
- ✓ Appearance of a blue or purple color indicates the presence of amino acids or proteins containing free amino groups.

12. Define primary, secondary, tertiary and quaternary structure of proteins.

Primary Structure: The primary structure of a protein is the linear sequence of amino acids joined together by peptide bonds.

Secondary Structure: The secondary structure is the regular folding of the polypeptide chain into forms such as alpha-helix and beta-pleated sheet due to hydrogen bonding.

Tertiary Structure: The tertiary structure is the three-dimensional shape of a protein formed by folding of the secondary structure through interactions among side chains of amino acids.

Quaternary Structure: The quaternary structure is the arrangement of two or more polypeptide chains that combine to form a functional protein. For example, hemoglobin consists of four polypeptide chains.

13. What is spot test? Why it is used?

A spot test is a simple chemical test in which a small drop of a sample is placed on filter paper, spot plate, or another surface and treated with specific reagents to observe a color change or reaction.

It is used for the quick detection and identification of substances such as proteins, amino acids, sugars, or ions. Spot tests require only a small amount of sample and are commonly used in laboratories for rapid analysis.

14. How will you prepare dilute iodine solution from stock iodine solution?

To prepare a dilute iodine solution from a stock iodine solution, first take a clean test tube or measuring flask. Measure a small quantity of the stock iodine solution using a pipette or measuring cylinder. Then add distilled water in a larger amount to reduce the concentration of the solution. For example, mix 1 mL of stock iodine solution with 9 mL of distilled water to make a dilute iodine solution. After adding water, shake or stir the mixture properly to ensure uniform dilution. This dilute iodine solution is commonly used in laboratory tests such as the starch test.

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15. Define concentration of solution and its units.

The concentration of a solution is the amount of solute dissolved in a specific amount of solvent or solution. It indicates how much solute is present in the solution. Common units of concentration include:

1. **Molarity (M):** Number of moles of solute per liter of solution.
2. **Molality (m):** Number of moles of solute per kilogram of solvent.
3. **Percentage (%):** Amount of solute present in 100 parts of solution.
4. **Normality (N):** Number of gram equivalents of solute per liter of solution.
5. **Parts per million (ppm):** Parts of solute per million parts of solution.

16. Explain lipids.

Lipids are a group of organic compounds that are insoluble in water but soluble in organic solvents such as ether, chloroform, and alcohol. They are mainly composed of carbon, hydrogen, and oxygen. Lipids include fats, oils, waxes, and steroids. They serve as a major source of energy in living organisms, provide insulation and protection to organs, and form an important part of cell membranes (phospholipids). Lipids also help in storing energy for long-term use.

17. Name the test we perform to detect the presence of proteins.

The presence of proteins is detected by the Biuret test**.**

18. How we locate the amino acids on chromatogram?

Amino acids are located on a chromatogram by developing the chromatogram using a suitable solvent and then spraying it with ninhydrin reagent. After spraying, the chromatogram is gently heated, and amino acids appear as colored spots (usually purple or blue). The position of these spots is then marked and compared with known standards to identify different amino acids based on their movement (Rf values).

19. Name reagents of iodine test.

The reagents used in the iodine test are:

1. Iodine solution (I_2 in potassium iodide, KI)
2. Distilled water (for dilution, if needed)
3. Sample solution (for testing presence of starch)

20. What is amylase? Where it is naturally found and how its activity can be determined?

Amylase:

Amylase is a digestive enzyme that breaks down starch into simpler sugars like maltose and glucose.

Natural occurrence:

It is naturally found in human saliva (salivary amylase), pancreatic juice (pancreatic amylase), and also in plants and some microorganisms.

Determination of activity:

Its activity can be determined by iodine test. When amylase acts on starch, the blue-black color of iodine disappears over time. The rate at which the color disappears indicates the enzyme activity.

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21. What are carbohydrates and their major types?

Carbohydrates:

Carbohydrates are organic compounds made of carbon, hydrogen, and oxygen, usually in the ratio 1:2:1. They are an important source of energy for living organisms.

Major types of carbohydrates:

1. ***Monosaccharides*:** Simple sugars like glucose and fructose.
2. ***Disaccharides*:** Formed by two monosaccharides, e.g., sucrose and lactose.
3. ***Polysaccharides*:** Complex carbohydrates like starch, cellulose, and glycogen.

22. Differentiate between Molarity, Molality and Normality.

Molarity (M):

Molarity is the number of moles of solute present in one liter (dm^3) of solution. It depends on the volume of the solution and changes with temperature.

Molality (m):

Molality is the number of moles of solute present in one kilogram of solvent. It does not change with temperature because it depends on mass, not volume.

Normality (N):

Normality is the number of gram equivalents of solute present in one liter of solution. It is mainly used in acid-base and redox reactions and depends on the nature of the reaction.

LONG QUESTIONS

1. Define chromatography. Write a note on a paper chromatography and retardation factor.

Chromatography:

Chromatography is a separation technique used to separate components of a mixture based on their movement through a stationary phase with the help of a mobile phase.

Paper Chromatography:

Paper chromatography is a type of chromatography in which a special paper acts as the stationary phase and a solvent acts as the mobile phase. A small spot of the sample is placed near the base of the paper, and the paper is placed in a solvent. The solvent moves upward by capillary action and carries different components of the mixture at different rates, resulting in separation.

Retardation Factor (Rf):

The retardation factor (Rf) is the ratio of the distance traveled by a solute to the distance traveled by the solvent front. It helps in identifying substances.

$$Rf = \frac{Distance\ traveled\ by\ solute}{Distance\ traveled\ by\ solvent}$$

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2. Explain Molisch's test.

Molisch's Test:

Molisch's test is a general and sensitive qualitative test used to detect the presence of all types of carbohydrates, including monosaccharides, disaccharides, and polysaccharides.

Principle:

In this test, carbohydrates are dehydrated by concentrated sulfuric acid (H_2SO_4) to form furfural (from pentoses) or hydroxymethylfurfural (from hexoses). These products then react with α-naphthol (Molisch reagent) to form a violet or purple colored complex at the interface of the two layers.

Procedure:

Take 2–3 mL of the test solution in a clean test tube. Add 2–3 drops of Molisch reagent (α-naphthol solution in alcohol) and mix gently. Then carefully add concentrated sulfuric acid along the sides of the test tube so that it forms a separate layer without mixing. A violet or purple ring at the junction of the two layers indicates a positive test for carbohydrates.

Result/Observation:

Formation of a purple/violet ring confirms the presence of carbohydrates in the sample.

3. Write principle of Benedict's test, elucidating the role of Sodium carbonate, copper sulfate, sodium citrate and reducing carbohydrates. Why the formation of red precipitates signify the positive result?

Principle of Benedict's Test:

Benedict's test is based on the ability of reducing carbohydrates (such as glucose, fructose, lactose, and maltose) to reduce copper(II) ions (Cu^{2+}) present in Benedict's reagent to copper(I) ions (Cu^{+}) under alkaline conditions when heated. This reduction leads to the formation of a colored precipitate.

Role of components:

- ***Copper sulfate ($CuSO_4$):*** Provides Cu^{2+} ions, which are reduced by reducing sugars.
- ***Sodium carbonate (Na_2CO_3):*** Provides alkaline medium necessary for the reaction.
- ***Sodium citrate:*** Acts as a complexing agent; it keeps Cu^{2+} ions in solution by preventing their precipitation as copper hydroxide in alkaline conditions.
- ***Reducing carbohydrates:*** Act as reducing agents; they donate electrons to Cu^{2+} ions and get oxidized to carboxylic acids.

Reason for red precipitate formation:

During the reaction, Cu^{2+} ions are reduced to Cu^{+} ions, which form insoluble copper(I) oxide (Cu_2O). This compound appears as a brick-red precipitate. The intensity of the red color depends on the amount of reducing sugar present, so a red precipitate indicates a positive result.

4. What do you know about proteins and its structure?

Proteins:

Proteins are large, complex biomolecules made up of amino acids joined together by peptide bonds. They contain carbon, hydrogen, oxygen, nitrogen, and sometimes sulfur. Proteins are essential for growth, repair of tissues, enzyme formation, hormones, and structural components of cells.

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Structure of Proteins:

Proteins have four levels of structural organization:

- ***Primary structure:*** Linear sequence of amino acids linked by peptide bonds.
- ***Secondary structure:*** Folding of polypeptide chain into α-helix or β-pleated sheet due to hydrogen bonding.
- ***Tertiary structure:*** Three-dimensional folding of the polypeptide chain due to interactions like hydrogen bonds, ionic bonds, hydrophobic interactions, and disulfide bonds.
- ***Quaternary structure:*** Arrangement of two or more polypeptide chains to form a functional protein, e.g., hemoglobin.

5. How will you prepare 10% diluted solution from 37% concentrated stock solution?

To prepare a 10% solution from 37% stock solution, use the dilution formula:

$$C1V1 = C2V2$$

Where:
C_1 = 37% (stock solution)
C_2 = 10% (required solution)

So,
$37 \times V_1 = 10 \times V_2$
Assume V_2 = 100 mL (final volume):
$V_1 = (10 \times 100) / 37 = 27.03$ mL

Procedure:

Take 27 mL of 37% stock solution and dilute it with distilled water up to 100 mL total volume. Mix well to obtain a uniform 10% solution.

6. How the enzyme activity does is affected with the change in temperature? Explain the procedure.

Effect of Temperature on Enzyme Activity:

Enzyme activity increases with rise in temperature up to an optimum temperature (usually around 37°C in humans). At low temperature, enzymes become less active due to reduced molecular movement. As temperature increases, reaction rate increases. Beyond the optimum temperature, enzyme activity decreases sharply because enzymes get denatured (loss of shape of active site), so substrate cannot bind properly.

Procedure (to study effect of temperature):

Take equal amounts of enzyme (e.g., amylase) and substrate (starch) in separate test tubes. Place different test tubes in water baths at different temperatures (e.g., 0°C, 25°C, 37°C, and 60°C). After incubation, mix enzyme and substrate and allow reaction to proceed for a fixed time. Then test each mixture using iodine solution. The tube where starch disappears fastest (no blue-black color) shows highest enzyme activity. This helps determine the effect of temperature on enzyme activity.

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7. Write procedure for solubility test.

Procedure for Solubility Test:

Take a clean and dry test tube and add a small, measured amount of the given solid substance (solute). Then add about 2–5 mL of distilled water (solvent) to the test tube using a measuring cylinder or dropper. Close the test tube with a stopper and shake it gently for 1–2 minutes to ensure proper mixing. Observe whether the substance dissolves completely, partially, or remains undissolved at the bottom.

If the solute dissolves completely and forms a clear solution, it is considered soluble. If only a part dissolves and the rest settles, it is partially soluble. If no visible dissolution occurs, it is insoluble. For further confirmation, the test may also be repeated at different temperatures, as solubility of many substances increases with temperature.

8. How will you prepare 5% NaCl solution? Also write its procedure.

Preparation of 5% NaCl Solution:

A 5% NaCl solution means 5 g of sodium chloride is present in 100 mL of solution.

Calculation:

5% (w/v) = 5 g NaCl in 100 mL solution.

Procedure:

Take a clean beaker and weigh 5 g of sodium chloride (NaCl) using a balance. Add a small amount of distilled water and stir it with a glass rod until the salt completely dissolves. Then transfer the solution into a 100 mL volumetric flask. Add more distilled water up to the 100 mL mark. Shake the flask well to ensure uniform mixing. The prepared solution is a 5% NaCl solution.

9. Write a note on importance of qualitative test of protein in the biochemical analysis.

Importance of Qualitative Test of Proteins in Biochemical Analysis:

Qualitative tests of proteins are used to detect the presence or absence of proteins in a given sample without measuring their exact amount. These tests are very important in biochemical analysis because they help in identifying protein-containing substances in food, biological fluids, and experimental samples. Common tests include the Biuret test, Ninhydrin test, and Xanthoproteic test.

These tests are useful in clinical diagnosis for detecting abnormal protein levels in urine or blood, which may indicate diseases such as kidney disorders or malnutrition. In food analysis, they help determine protein content in different foods. In research laboratories, qualitative protein tests are used to confirm the presence of proteins before performing further quantitative analysis. Thus, these tests are essential for medical, nutritional, and biochemical studies.

10. Write method for Lipid extraction.

Method for Lipid Extraction (Procedure):

- ✓ Take a known quantity of biological sample (e.g., crushed seeds, tissue, or food sample).
- ✓ Grind the sample thoroughly using a mortar and pestle to break cell structures.
- ✓ Transfer the ground material into a clean test tube or conical flask.
- ✓ Add an appropriate organic solvent such as ethanol, ether, or chloroform to the sample.
- ✓ Shake the mixture well for 5–10 minutes to allow lipids to dissolve in the solvent.
- ✓ Leave the mixture for some time so complete extraction occurs.
- ✓ Filter the mixture using filter paper to separate solid residues from the liquid extract.
- ✓ Collect the filtrate containing dissolved lipids.
- ✓ Evaporate the solvent gently by air drying or mild heating.
- ✓ The remaining oily or greasy residue confirms the presence of lipids.

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11. Name five glass wares used in lab.

Five commonly used laboratory glass wares are:

1. Beaker
2. Test tube
3. Conical flask (Erlenmeyer flask)
4. Measuring cylinder
5. Pipette

12. Explain the effect of concentration on enzyme activity.

Effect of Concentration on Enzyme Activity:

Enzyme activity is affected by both enzyme concentration and substrate concentration.

When enzyme concentration increases, the reaction rate increases because more active sites are available for substrate binding, provided that substrate is available in excess. However, after a certain point, the rate becomes constant if substrate becomes limiting.

When substrate concentration increases, the reaction rate also increases because more substrate molecules collide with enzyme active sites. Eventually, all active sites become saturated, and the reaction reaches a maximum rate (Vmax), where further increase in substrate does not increase activity.

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